Thermodynamics

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Thermodynamic System

Thermodynamics is the study of exchange of heat energy between bodies and inter-conversion between heat energy and mechanical energy.

Thermodynamic system & Surrounding

A thermodynamic system is a confined space with certain boundaries having large number of atoms or molecules such that it has certain value of pressure, volume & temperature and on which the effect of these variables is to be studied.
Anything outside the thermodynamics system which exchanges matter or energy with the system and tends to change the properties of the system is known as its Surrounding.

Types of thermodynamics systems

Taking into consideration the type of interaction between a system and its surrounding, thermodynamics systems can be classified into these types —

a. **Open system:** It can exchange both energy and matter with its surroundings.

   e.g., boiling water without a lid.

b. **Closed system:** It can exchange only energy with its surrounding.

   e.g., a pressure cooker.

c. **Isolated system:** A system in which there can be no exchange of matter and energy with its surrounding.

   e.g., a thermos flask.

Universe

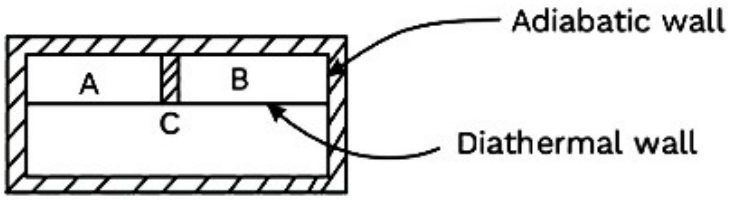
The system and its surrounding are together known as the universe.

Thermal equilibrium

Thermal equilibrium is a condition in which net heat exchange between two systems which are in thermal contact with each other is zero.

Zeroth law of thermodynamics

If system A and B are separately in thermal equilibrium with a third system C, then system A and system B are in thermal equilibrium with each other and exist at the same temperature. It is Zeroth law of thermodynamics which introduced the concept of temperature.



Thermodynamic state variables

Thermodynamic state for a system is the condition of the system at a specific time. Thermodynamic state variables are the macroscopic quantities which determine the thermodynamic state of a system. They include pressure (P), temperature (T), Volume(V), composition (n), internal energy (U) etc.

Equation of thermodynamic state

An equation relating the thermodynamic state variables is called equation of state. For an ideal gas, state equation is given by

$$PV = nRT = NKT$$

Where, $n \rightarrow$ number of moles of gas

$N \rightarrow$ number of atoms or molecules

$R \rightarrow$ Gas constant (8.314 J/Kmol)

$K \rightarrow$ Boltzmann constant (1.38×10^{-23} J/K)

Q. **One mole each of hydrogen, oxygen and nitrogen are mixed in a closed container of volume 10 litres and temperature 27° C. Calculate the pressure exerted by the mixture. (R = 8.314 J mol^{-1} k^{-1})**

Sol. According to Dalton's law of partial pressure

$$P = P_1 + P_2 + P_3$$

$$P = \frac{nRT}{V} + \frac{nRT}{V} + \frac{nRT}{V} = \frac{3nRT}{V}$$

Where, n = 1 mole

$$\therefore \quad P = \frac{3RT}{V} = \frac{3 \times 8.314 \times 300}{10 \times 10^{-3}}$$

$$= 7.48 \times 10^5 \, Nm^{-2}$$

Q. **A cylinder of cross—sectional area A has two pistons of negligible mass separated by distance l loaded with spring of negligible mass. An ideal gas at temperature T_1 is in the cylinder where the springs are relaxed. When the gas is heated by some means, its temperature becomes T_2 and the spring gets compressed by $\frac{l}{2}$ each. If P_0 is atmospheric pressure and spring constant $k = \frac{2P_0A}{l}$, then find the ratio of T_2 and T_1.**



Sol. By equation of thermodynamic state, $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

Where,

$V_1 = \ell A$ and $V_2 = 2\ell A$ and $P_1 = P_0$ and $P_2 = \frac{kx}{A} + P_0 = 2P_0 \Rightarrow \frac{T_2}{T_1} = \frac{P_2V_2}{P_1V_1} = 4$

Q. **A vessel is containing 1 g of oxygen at a pressure of 10 atm and a temperature of 47° C. The pressure in the vessel becomes 5/8 th of its initial value due to some leakage and the temperature falls to 27° C. Find the volume of the vessel and the mass of oxygen that is leaked out.**

Sol. The pressure, temperature, and the number of moles of oxygen in the vessel change due to leak while the volume remains fixed. Hence using PV = nRT, we have

$$\frac{P_1}{n_1T_1} = \frac{P_2}{n_2T_2} \Rightarrow n_2 = \frac{T_1}{T_2} \times \frac{P_2}{P_1} \times n_1 \quad \ldots\left(n_1 = \frac{1\,g}{32\,g} = \frac{1}{32}\right)$$

Hence, $n_2 = \frac{320}{300} \times \frac{5}{8} \times \frac{1}{32} = \frac{1}{48}$ moles

Mass leaking out $= 1\,g - n_2(32g) = \frac{1}{3}g$

Volume of vessel, $V = \frac{n_1RT_1}{P_1}$

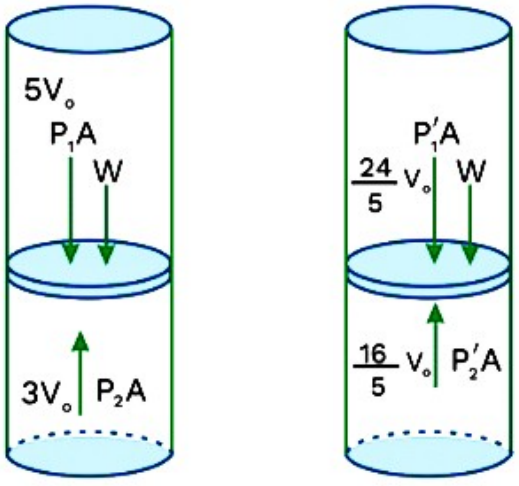
$$\Rightarrow V = \left(\frac{1}{32}\right)\frac{8.3 \times 320}{10 \times 1 \times 10^5} = 8.3 \times 10^{-5}\,m^3 = 0.083\,L$$

Q. **A vertical cylinder contains a movable frictionless piston separating the former into two portions. The two portions are filled with ideal gases, and at equilibrium, the ratio of the volume of the upper portion to that of the lower portion is 5: 3. When the temperature of the system is doubled, the ratio becomes 3: 2. Determine the ratio of number of molecules in the upper portion to that of the lower portion.**

Sol. Let the weight of the piston be W and cross—sectional area of the cylinder be A. Let the initial and the final pressures of the upper and the lower regions be P_1 and P_2 respectively. Further, if initial temperature be taken to be T_0, then final temperature will be $2T_0$.

Let the volume of upper part is $5V_0$ and lower part is $3V_0$, hence total volume of the cylinder is $(5V_0 + 3V_0 = 8V_0)$. When the cylinder is heated to $2T_0$, the ratio of volume of upper and lower part becomes 3:2. Hence, the volume of upper part $V_1' = \frac{3}{5} \times 8V_0 = \frac{24}{5} V_0$ and volume of lower part $V_2' = \frac{2}{5} \times 8V_0 = \frac{16}{5} V_0$.

From the FBD of the piston, initially.



At temperature T_0 At temperature $2T_0$

$$P_1A + W = P_2A$$

$$\text{or } \frac{n_1 RT_0}{5V_0} + \frac{W}{A} = \frac{n_2 RT_0}{3V_0} \quad \text{..... (i)}$$

where n_1 and n_2 are the numbers of moles of gases in the two portions.

Finally, from the FBD of the piston,

$$P_1 'A + W = P_2 'A$$

or $$\frac{n_1 R(2T_0)}{3(8V_0/5)} + \frac{W}{A} = \frac{n_2 R(2T_0)}{2(8V_0/5)} \quad \text{..... (ii)}$$

Subtracting equation (ii) from (i), we have

$$n_1\left(\frac{1}{5} - \frac{5}{12}\right) = n_2\left(\frac{1}{3} - \frac{5}{8}\right)$$

or, $n_1 : n_2 = 35:26$

First Law of Thermodynamics

Introduction to Heat, Internal energy and Work done

Heat: Heat is the form of energy that is transferred from one system to another as a result of temperature difference between them.

Heat flows from high temperature system to the low-temperature system.

Heat is a path function.

Internal energy: The internal energy of a system is the sum of internal kinetic energy due to molecular motion and internal potential energy due to intermolecular forces. For ideal gas, intermolecular forces are zero and hence its potential energy is also zero. So, the internal energy for ideal gas is only due to internal kinetic energy which is a function of absolute temperature (T) given as

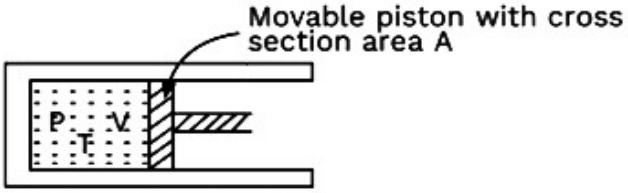
$$U = n\frac{f}{2}RT$$

Change in internal energy, $\Delta U = n\frac{f}{2}R\Delta T$

Internal energy is a point function.

Work done by a gas:

Let P and V be the pressure and volume of a gas in a cylinder with a movable piston as shown in fig.



Then, force exerted by the gas on the piston is, $F = P \times A$

When the piston is moved outward through a small distance dx, then the work done by the gas is, $dW = F\,dx$

$\Rightarrow$ For a displacement which changes the volume from V_i to V_f, $W = \int_{V_i}^{V_f} P\,dV$

Work is a path function,

Sign conventions:

1. Heat gained by a system $\rightarrow$ '+ ve', heat lost by a system $\rightarrow$ '– ve'
2. Increase in internal energy of the system $\rightarrow$ '+ ve'
3. Work done by a system $\rightarrow$ '+ ve' , Work done on a system $\rightarrow$ '– ve'

Thermodynamic Process:

The process of change in the state of a thermodynamic system is termed as thermodynamics process.

Indicator Diagram:

A graph between any two of the three thermodynamic variables P, V and T is known as indicator diagram,

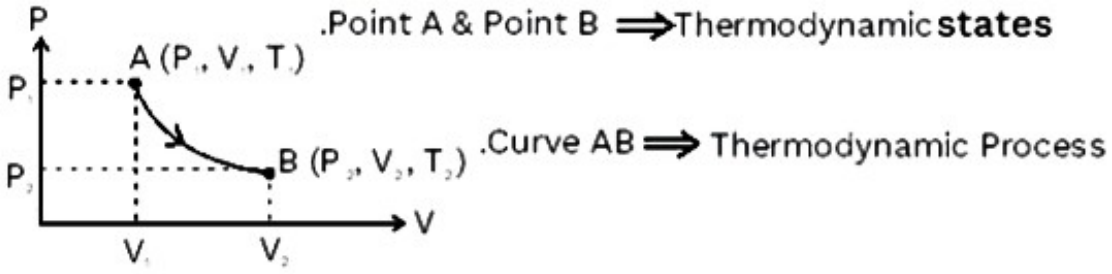
Types of indicator diagram —

(a) P—V graph

(b) V—T graph

(c) P—T graph

Two thermodynamics variables are sufficient to describe the behaviour of a system.
Every point of indicator diagram represents a unique thermodynamic state.
Every curve on indicator diagram represents a unique thermodynamic process.



P—V Diagram

Each curve on P—V diagram has a characteristic function between P and V as P = f(V), known as equation of the thermodynamic process.

It is already discussed, the work done by a gas is given as

$$W = \int_{V_1}^{V_2} P\,dV$$

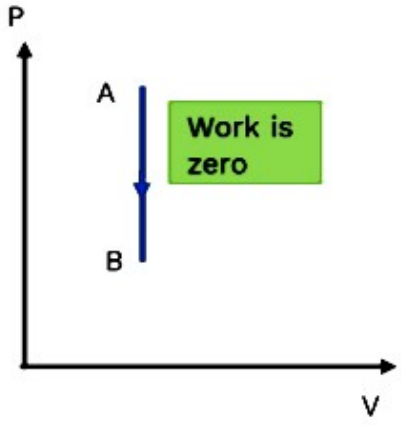
And if equation of thermodynamic process is known, then

$$W = \int_{V_1}^{V_2} f(V).dV$$

Hence the work done by a gas is equal to the area under P—V graph.

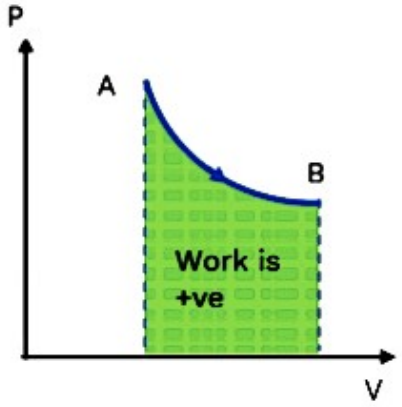
Different possible cases are given below-

a. If volume is constant during a thermodynamic process, then, W=0.



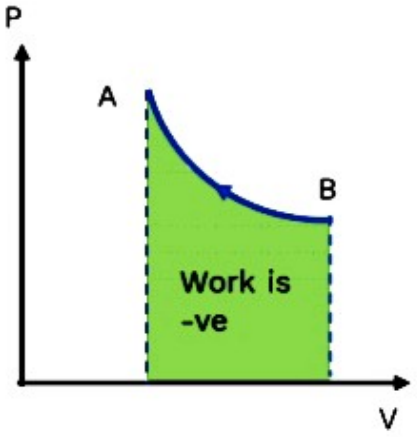
b. If volume is increasing during a process, then dV → '+ ve'

⇒ W_{AB} → '+ ve'



c. If volume is decreasing during a process, then dV → '– ve'

⇒ W_{AB} → '– ve'



First Law of Thermodynamics:

The first law of thermodynamics is basically the law of conservation of energy.

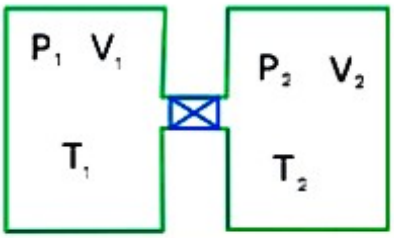
If heat is transferred to a system capable of doing work, then a part of this heat is used to do work against surrounding and the other part increases the internal energy of the system.

Mathematical statement, $\delta Q = \delta W + dU$ or

$Q = W + \Delta U$

This law is applicable to all the three phases of matter i.e., solid, liquid and gas.

Q. **Two non-conducting containers having volume V_1 and V_2 contain monoatomic and diatomic gases, respectively. They are connected as shown in figure. Pressure and temperature in the two containers are P_1, T_1 and P_2, T_2 respectively. Initially stop cock is closed, if the stop cock is opened, find the final pressure and temperature:**



Sol. $n_1 = \frac{P_1V_1}{RT_1}$ $\qquad$ $n_2 = \frac{P_2V_2}{RT_2}$

$n = n_1 + n_2$ (Number of moles are conserved)

Finally, pressure in both parts & temperature of the both the gases will become equal.

$$\frac{P(V_1 + V_2)}{RT} = \frac{P_1V_1}{RT_1} + \frac{P_2V_2}{RT_2}$$

From energy conservation,

$$\frac{3}{2}n_1RT_1 + \frac{5}{2}n_2RT_2 = \frac{3}{2}n_1RT + \frac{5}{2}n_2RT$$

$$\Rightarrow T = \frac{(3P_1V_1 + 5P_2V_2)T_1T_2}{3P_1V_1T_2 + 5P_3V_2T_1} \quad \Rightarrow P = \left(\frac{3P_1V_1 + 5P_2V_2}{3P_1V_1T_2 + 5P_2V_2T_1}\right)\left(\frac{P_1V_1T_2 + P_2V_2T_1}{V_1 + V_2}\right)$$

Q. **An ideal gas in a cylindrical vessel is confined by a piston at a constant pressure of 10^5 Pa. When 2×10^4 J of heat is added to it, the volume of gas expands from 0.15 m^3 to 0.25 m^3. (a) What is the work done by the system in this process. (b) What is the change in internal energy of the system.**

Sol. (a) We know that work done by a gas in a process is given as

$$W = \int_{V_1}^{V_2} PdV = P(V_2 - V_1)$$

$$= 10^5 [0.25 - 0.15] = 10^4 \text{ J}$$

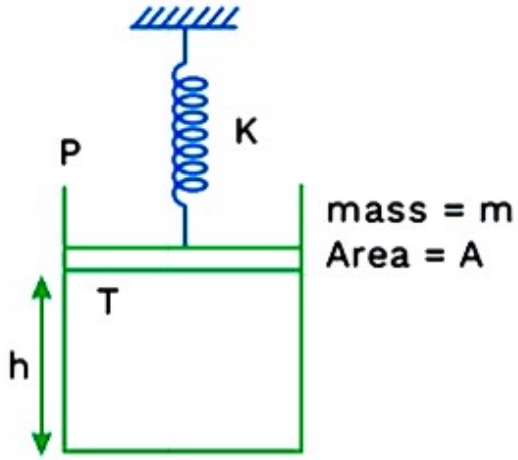
(b) From first law of thermodynamics, in the process we have

$$Q = \Delta U + W$$

Here $2 \times 10^4 = \Delta U + 10^4$

Or $\Delta U = 10^4 \text{ J}$.

Q. **A non-conducting piston of mass m and area of cross section A is placed on a non-conducting cylinder as shown in figure. Temperature, spring constant, height of the piston are given by T, K, h respectively. Initially spring is relaxed, and piston is at rest. Find:**
(i) Number of moles
(ii) Work done by gas to displace the piston by distance d when the gas is heated slowly.

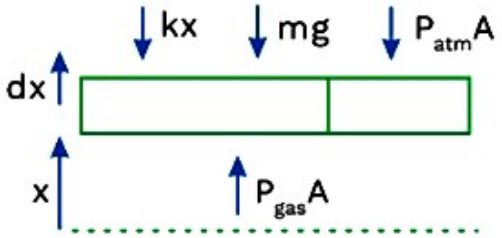


Sol. (i) $PV = nRT$

$$\Rightarrow \quad \left(P_{atm} + \frac{mg}{A}\right) Ah = nRT$$

$$\Rightarrow \quad n = \frac{\left(P_{atm} + \frac{mg}{A}\right) Ah}{RT}$$

(ii)



Applying Newton's law on the piston

$mg + P_{atm} A + kx = P_{gas} A$

$$W_{gas} = \int_0^d P_{gas} A \, dx$$

$$= \int_0^d (mg + P_{atm}A + Kx)\, dx = mgd + P_{atm}Ad + \frac{1}{2}Kd^2$$

Q. **1 gm water at 100° C is heated to convert into steam at 100° C, 1 atm. Find out change in internal energy of water. It is given that volume of 1 gm water at 100° C = 1 cc, volume of 1 gm steam at 100° C = 1671 cc. Latent heat of vaporization = 540 cal/g.**
(Mechanical equivalent of heat J = 4.2 J/cal.)

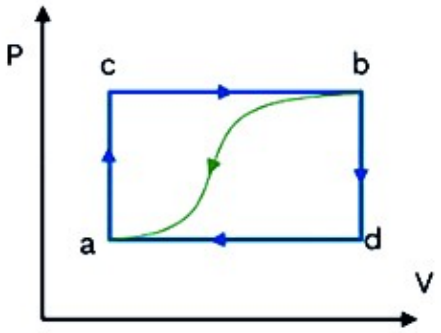
Sol. From first law of thermodynamics, $Q = \Delta U + W$

$Q = mL = 1 \times 540\,\text{cal.} = 540\,\text{cal.}$

$$W = P\Delta V = \frac{10^5 (1671 - 1) \times 10^{-6}}{4.2} = \frac{10^5 \times (1670) \times 10^{-6}}{4.2} = 40\,\text{cal.}$$

$\Delta U = 540 - 40 = 500\,\text{cal.}$

Q. **When a system is taken from state a to state b, in figure along the path $a \rightarrow c \rightarrow b$, 60 J of heat flows into the system and 30 J of work is done:**



(i) How much heat flows into the system along the path $a \rightarrow c \rightarrow b$ if the work is 10 J.

(ii) When the system is returned from b to a along the curved path, the work done by the system is —20 J. Does the system absorb or liberate heat and how much?

Sol. (i) For the path a- c- b, $\Delta U = Q - W = 60 - 30 = 30\,J$ or $U_b - U_a = 30\,J$

Along the path a- c-b if work done is 10 J, then, $Q = \Delta U + W = 30 + 10 = 40\,J$.

(ii) Along the curved path b- a, $Q = (U_a - U_b) + W = (-30) + (-20) = -50\,J$,

As heat is negative, so heat flows out of the system.

Q. **Gaseous hydrogen initially at STP in a container of volume $5 \times 10^{-3}\ m^3$ is cooled by 55 K. Find the change in internal energy and amount of heat lost by the gas.**

Sol. Initially in standard conditions, the gas pressure is

$P = 10^5\ Pa$

Gas temperature is $T = 273\ K$

Gas volume is $V = 5 \times 10^{-3}\ m^3$

If n moles of gas are there, then from gas law, we have

$PV = nRT$

Or $n = \frac{PV}{RT} = \frac{10^5 \times 5 \times 10^{-3}}{8.314 \times 273} = 0.22$ mol.

Thus, change in internal energy in a process is given as:

$$\Delta U = \frac{f}{2} nR\Delta T = \frac{5}{2} \times 0.22 \times 8.314 \times 55 = 251.5\,J$$

As gas is enclosed in a container, its volume remains constant during the process, thus work done in this process is zero and according to first law of thermodynamics, we have

$$Q = \Delta U = -251.5\,J$$

HEAT CAPACITY:

The amount of heat required to raise the temperature of a gas by one unit, is known as heat capacity.

If Q amount of heat is given to a gas and its temperatures raises by ΔT, then heat capacity of gas is ,

$$S = \frac{Q}{\Delta T} \ JK^{-1}$$

SPECIFIC HEAT CAPACITY:

The amount of heat required for a unit mass of a gas to raise its temperature by one unit, is known as specific heat capacity.

If Q amount of heat is given to a mass m of the gas and its temperature raises by ΔT, then specific heat capacity is,

$$s = \frac{Q}{m\Delta T} \ J/kg.K \quad \text{or} \quad J\,kg^{-1}K^{-1}$$

MOLAR HEAT CAPACITY:

The amount of heat required for one mole of a gas to raise its temperature by one unit, is known as molar heat capacity.

If Q amount of heat is given to n moles of a gas and its temperature raises by ΔT, then molar heat capacity of gas is

$$C = \frac{Q}{n\Delta T} \ J\,mol^{-1}K^{-1}$$

Note:

1. Relation between heat capacity (S), specific heat capacity (s) and molar heat capacity (C) of a gas is given as

   $$S = ms = nC$$

2. At room temperature, average molar heat capacity of all metals is same and is nearly equal to 3R (6 cal. $mol^{-1}K^{-1}$).

   But for gases these properties depend on the process of heating. For different types of processes of heating, there will be different values of molar/specific heat capacity for the same gas.

   Mayer's Equation: $C_p - C_v = R$ (for ideal gases only)

3. As a characteristic property, a gas has two molar heat capacities, these are –

   (a) Moles heat capacity at constant volume (C_v)

   (b) Moles heat capacity at constant volume (C_p)

MOLAR HEAT CAPACITY AT CONSTANT VOLUME (C_v):

It is the amount of heat required for one mole of a gas to raise its temperature by one unit at constant volume.

For a gas, $C_v = \frac{f}{2}R$ where f → degree of freedom

For a mixture of non – reacting gases,

$$C_V = \frac{n_1C_{V_1} + n_2C_{V_2} + ------ + n_NC_{V_N}}{n_1 + n_2 + ------ + n_N}$$

MOLAR HEAT CAPACITY AT CONSTANT PRESSURE (C_p):

It is the amount of heat required for one mole of a gas to raise its temperature by one unit at constant pressure.

For a gas, $C_p = \left(\frac{f}{2} + 1\right)R$

For a mixture of non-reacting gases,

$$C_P = \frac{n_1C_{P_1} + n_2C_{P_2} + ------ + n_NC_{P_N}}{n_1 + n_2 + ------ + n_N}$$

RATIO OF HEAT CAPACITIES OF A GAS (γ):

The ratio of heat capacities is also a characteristic property of a gas and it is also termed as adiabatic exponent of a gas.

For a gas, $\gamma = \frac{C_P}{C_V} = \left(\frac{f+2}{f}\right)$

Some important relations,

$$f = \left(\frac{2}{\gamma - 1}\right),\ C_V = \frac{R}{\gamma - 1}$$

$$\text{and } C_P = \frac{\gamma R}{\gamma - 1}$$

For a mixture of non-reacting gases,

$$\gamma = \frac{n_1C_{P_1} + n_2C_{P_2} + ------ + n_NC_{P_N}}{n_1C_{V_1} + n_2C_{V_2} + ------ + n_NC_{V_N}}$$

If $\gamma_1, \gamma_2, \gamma_3, ...\gamma_n$ are the ratio of specific heats for independent gases of a mixture, then As we know,

$$n_1C_{V_1}T + n_2C_{V_2}T + \text{- - - - - - - - } + n_NC_{V_N}T = (n_1 + n_2 + n_3 \text{- - - } n_N)C_{V_{eq}}T$$

$$\Rightarrow \frac{n_1RT}{(\gamma_1 - 1)} + \frac{n_2RT}{(\gamma_2 - 1)} + \text{- - - - - } + \frac{n_NRT}{(\gamma_N - 1)} = (n_1 + n_2 + \text{- - - - - } n_N)\frac{RT}{(\gamma_{eq} - 1)}$$

$$\Rightarrow \left[\left(\frac{n_1}{\gamma_1 - 1}\right) + \left(\frac{n_2}{\gamma_2 - 1}\right) + \text{- - - - - - - } + \left(\frac{n_N}{\gamma_N - 1}\right) = \frac{n_1 + n_2 + \text{- - - - - - - } + n_N}{(\gamma_{eq} - 1)}\right]$$

Thermodynamic Processes

There can be infinite types of thermodynamic processes but some of these are important. We will discuss in detail each of the important thermodynamic processes listed below-

a. Isothermal process
b. Isochoric process
c. Isobaric process
d. Adiabatic process
e. Polytropic process
f. Free expansion
g. Cyclic processes

(i) ISOTHERMAL PROCESS:

The temperature of the system remains constant during the isothermal process.

$T = \text{constant} \Rightarrow \Delta T = 0$

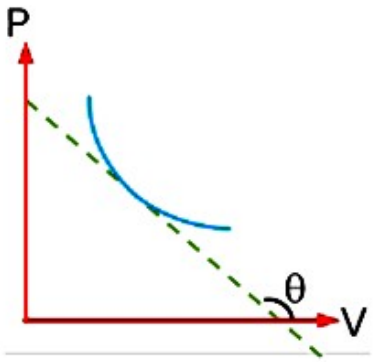
By equation of state for ideal gas, we get

$PV = C$, where $C(\text{Constant}) = nRT$

$\Rightarrow \left[P(V) = \frac{C}{V}\right]$ equation of thermodynamic process.

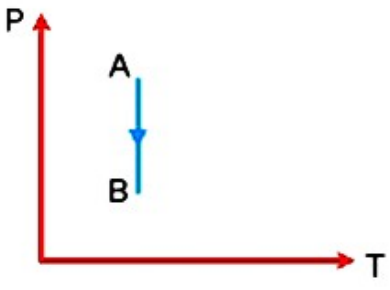
Indicator diagrams for an isothermal process.

a) P-V Graph



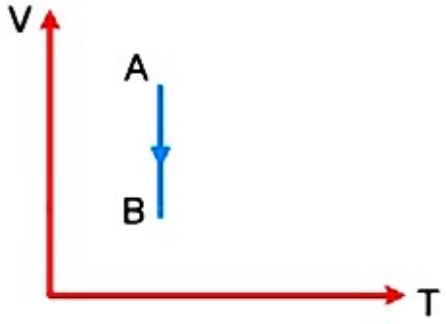
Slope of P–V curve at any point, $\tan\theta = \frac{dP}{dV} = -\frac{P}{V}$

b) P–T graph



Slope of P–T graph, $\frac{dP}{dT} = \infty$

c) V – T graph



Slope of V–T graph, $\frac{dV}{dT} = \propto$

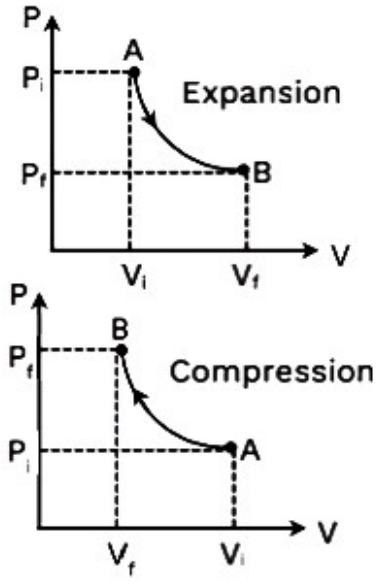
Work done in isothermal process

$$W = \int_{V_i}^{V_f} PdV = C\int_{V_i}^{V_f} \frac{1}{V} dV = nRT\,\ell n\left[\frac{V_f}{V_i}\right]$$

In terms of pressure, $W = nRT\,\ell n\left[\frac{P_i}{P_f}\right]$

$$\left.\begin{array}{l}\text{If } V_f > V_i \\ \text{or } P_i > P_f\end{array}\right] \text{ then W is positive.}$$

$$\left.\begin{array}{l}\text{If } V_f < V_i \\ \text{or } P_f > P_i\end{array}\right] \text{ then W is negative.}$$



Change in internal energy is zero as $\Delta T = 0$.
$\Rightarrow \Delta U = 0$
By first law of thermodynamics,
Heat, $Q = W + \Delta U$
Since $\Delta U = 0 \Rightarrow Q = W$

Above expression indicates that whole of the heat energy supplied to the system is utilized in doing external work.
For an isothermal process, system should be compressed or expanded very slowly so that there is sufficient time for heat exchange to keep the temperature constant.
Melting and boiling are isothermal processes.
The curves drawn for the same gas at different temperatures are mutually parallel.

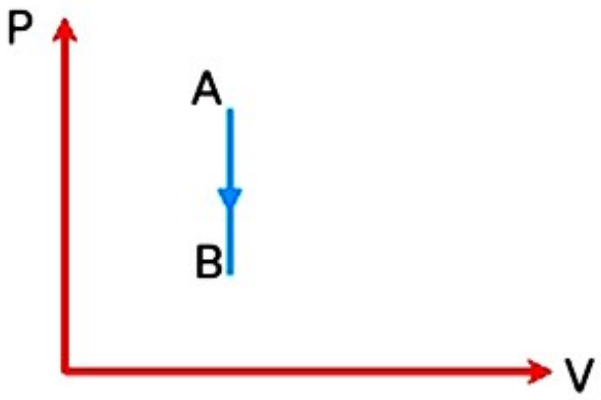
(ii) ISOCHORIC OR ISOMETRIC PROCESS:

Isochoric process takes place at constant volume of the system.

$V = \text{constant} \Rightarrow \Delta V = 0$

Equation of state, $\frac{P}{T} = \text{constant (say c)}$

Indicator diagrams of isochoric process

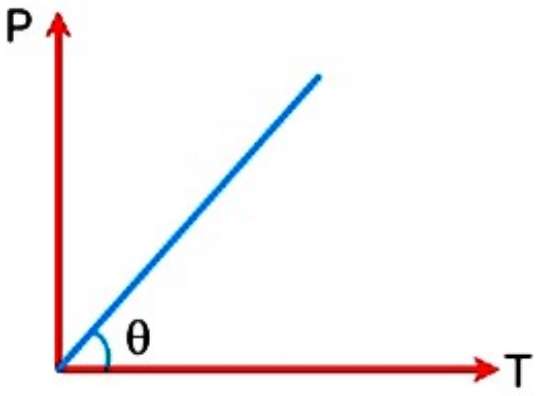


(a) P–V graph

Slope of P–V curve,

$$\tan\theta = \frac{dP}{dV} = \infty$$

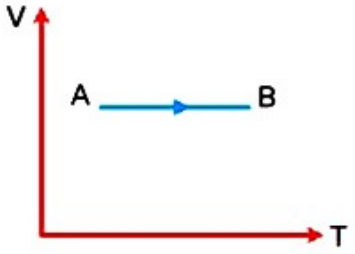
(b) P–T graph



Slope of P–T curve

$$\tan\theta = \frac{dP}{dT} = C$$

(c) V – T graph



Slope of V – T curve

$$\tan\theta = \frac{dV}{dT} = 0$$

Work done in isochoric process, $W = \int_{V_1}^{V_2} PdV = 0$

Change in internal energy is isochoric process,

$$\Delta U = n\frac{f}{2}R\Delta T$$

By first law of thermodynamics,

$$Q = W + \Delta U$$

Since $W = 0 \Rightarrow Q = \Delta U$

This expression indicates that whole of the heat supplied to the system is utilized for change in its internal energy.

Molar heat capacity at constant volume (C_v):

We know at constant volume,

$$Q = \Delta U$$

$$\Rightarrow nC_v\Delta T = n\frac{f}{2}R\Delta T$$

So, $C_v = \frac{f}{2}R$

Q. **There are two vessels. Each of them contains one mole of a monatomic ideal gas. Initial volume of the gas in each vessel is 8.3×10^{-3} m³ at 27° C. Equal amount of heat is supplied to each vessel. In one of the vessels, the volume of the gas is doubled without change in its internal energy, whereas the volume of the gas is held constant in the second vessel. The vessels are now connected to allow free mixing of the gas.**
Find the final temperature and pressure of the combined gas system.

Sol. According to first law of thermodynamics,

$$Q = \Delta U + W$$

For vessel 1, $\Delta U = 0$ [As no change in temperature]

or $Q = W$

or $Q = \int_{V_1}^{V_2} P\,dV$

For second vessel, $W = 0$ (as volume is constant)

Thus, heat supplied is given as

$$Q = nC_v\Delta T = n\left(\frac{3}{2}R\right)\Delta T$$

[As for monoatomic gas $C_v = \frac{3}{2}R$]

We get,

$$C_v = \frac{3}{2}R$$

or $$\Delta T = \frac{2}{3} \times 300 \times 0.693 = 138.6\,K$$

This is the change in the temperature of the gas of second vessel.

Now the final temperature of the gas in second vessel is

$$T' = T + \Delta T = 300 + 138.6 = 438.6\,K$$

Let after mixing, T_f and P_f be the final temperature and pressure, respectively.

Then as moles are equal, we have

$$T_f = \frac{T + (T + \Delta T)}{2} = \frac{300 + 438.6}{2} = 369.3K$$

For a gas system from gas law, we have

$$P_fV_f = nRT_f$$

or $$P_f = \frac{nRT_f}{V_f} = \frac{2 \times 8.3 \times 369.3}{2 \times 8.3 \times 10^{-3} + 8.3 \times 10^{-3}}$$

$$= 2.46 \times 10^5\ N/m^2$$

(iii) ISOBARIC PROCESS:

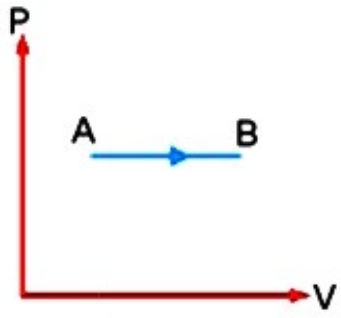
Pressure remains constant, but volume and temperature vary for change in state of the system,

$P = \text{constant} \quad \Rightarrow \Delta P = 0$

By equation of state, $\frac{V}{T} = \text{constant (say C)}$

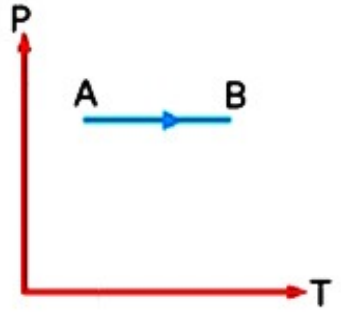
Indicator diagrams of isobaric process,

(a) P–V graph



Slope of P–V curve, $\tan\theta = \frac{dP}{dV} = 0$

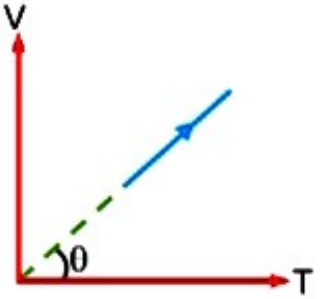
(b) P–T graph



Slope of P–T curve,

$$\tan\theta = \frac{dP}{dT} = 0$$

(c) V–T graph



Slope of V–T curve,

$$\tan\theta = \frac{dV}{dT} = C$$

Work done in isobaric process,

$$W = P\int_{V_i}^{V_f} dV = P(V_f - V_i) = nR\Delta T$$

Change in internal energy,

$$\Delta U = n\frac{f}{2}R\Delta T$$

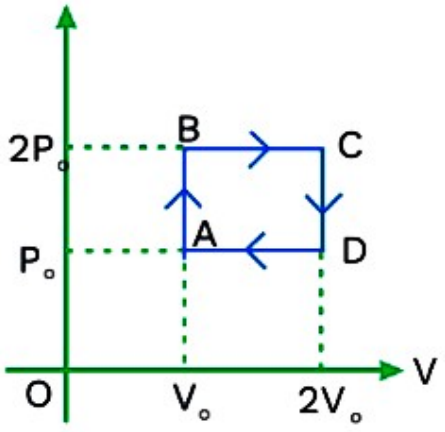
Molar heat capacity at constant pressure (C_p):
By first law of thermodynamics,
$Q = W + \Delta U$

$$\Rightarrow nC_p\Delta T = nR\Delta T + n\frac{f}{2}R\Delta T$$

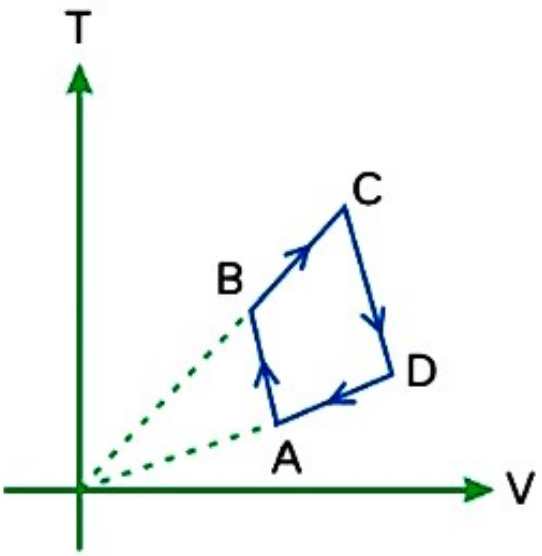
$$\Rightarrow nC_p\Delta T = n\left(\frac{f}{2}+1\right)R\Delta T$$

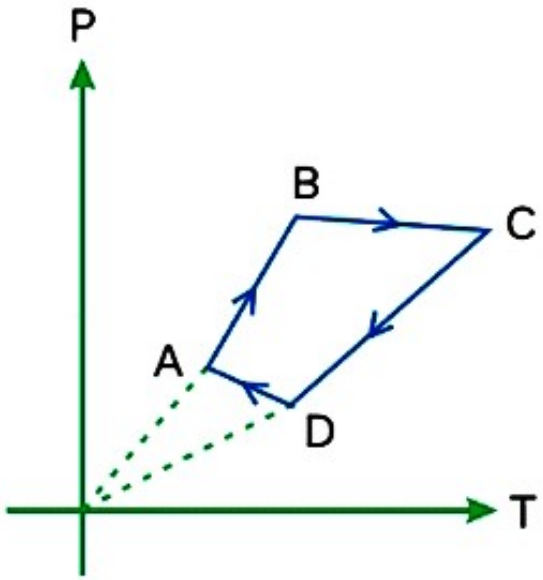
$$\text{So, } C_p = \left(\frac{f}{2}+1\right)R$$

Q. **Draw the corresponding T—V curve and P—T curve.**

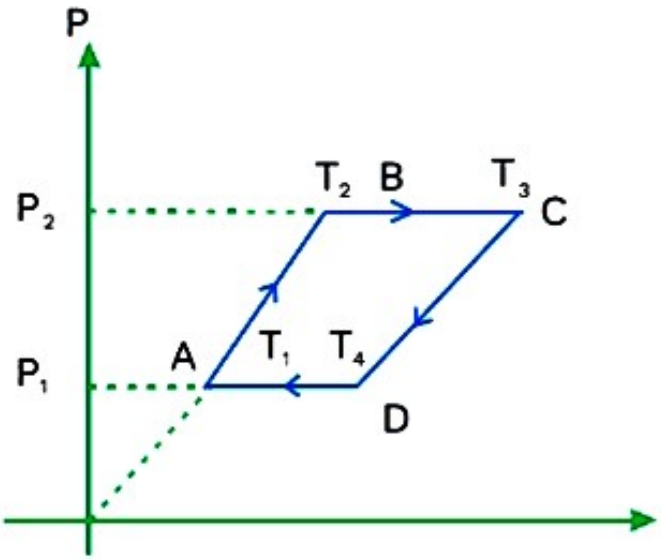


Sol. Line AB and CD are isochoric lines, line BC and DA are isobaric lines.
$\therefore$ The T-V curve and P-T curve are drawn as shown.





Q. **P—T curve of a cyclic process is shown. Find out the work done by the gas in the given process if number of moles of the gas are n.**



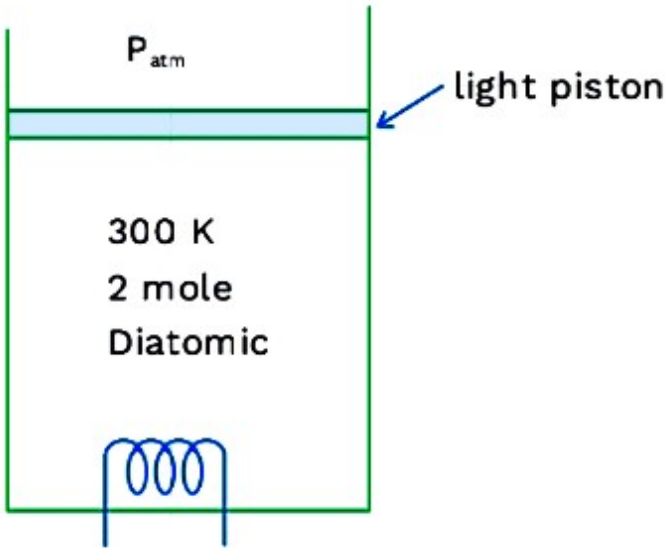
Sol. Since path AB and CD are isochoric therefore work done during AB and CD is zero. Path BC and DA are isobaric.

Hence $W_{BC} = nR\Delta T = nR(T_3 - T_2)$

$$W_{DA} = nR(T_1 - T_4)$$

Total work done $W_{BC} + W_{DA} = nR(T_1 + T_3 - T_4 - T_2)$

Q. **Two moles of a diatomic gas at 300 K are enclosed in a cylinder as shown in figure. Piston is light. Find out the heat given if the gas is slowly heated to 400 K in the following cases.**



(i) Piston is free to move
(ii) If piston does not move

Sol. (i) Since pressure is constant

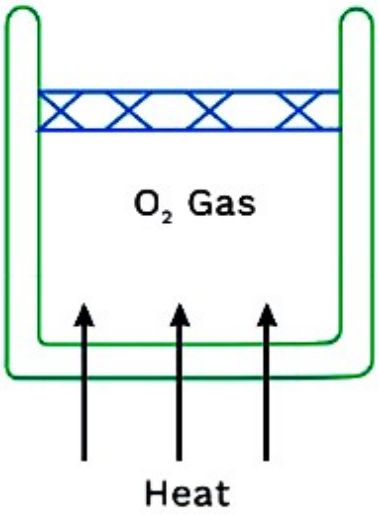
$$\therefore Q = nC_p\Delta T = 2 \times \frac{7}{2} \times R \times (400 - 300) = 700R$$

(ii) Since volume is constant

$\therefore W = 0$ and $Q = \Delta U$ (from first law)

$$Q = \Delta U = nC_v\Delta T = 2 \times \frac{5}{2} \times R(400 - 300) = 500R$$

Q. **Figure shows a cylindrical container containing oxygen gas and closed by a piston of mass 50 kg. Piston can slide smoothly in the cylinder. Its cross-sectional area is 100 cm^2 and atmospheric pressure is 10^5 Pa. Some heat is supplied to the cylinder so that the piston is slowly displaced up by 20 cm. Find the amount of heat supplied to the gas:**



Sol. As here position is open to atmosphere, the gas is under a constant pressure given as

$$P_{gas} = P_{atm} + \frac{Mg}{S} = 10^5 + \frac{50 \times 10}{100 \times 10^{-4}} Pa$$

$$= 10^5 + 5 \times 10^4 = 1.5 \times 10^5 Pa$$

It is given that the piston moves out by 20 cm thus increment is its volume is given as

$$\Delta V = S\Delta x = 100 \times 10^{-4} \times 20 \times 10^{-2} m^3 = 2 \times 10^{-3} m^3$$

Thus, in the process work done by gas is

$$W = P_{gas}\Delta V \qquad [\text{As } P_{gas} = \text{constant}]$$

$$= 1.5 \times 10^5 \times 2 \times 10^{-3} = 300 \text{ J}$$

For a process is which gas pressure is constant, work done can also be given as

$$W = nR\Delta T = P_{gas}\Delta V$$

or $nR\Delta T = 300\,J$

As in the process, gas pressure is constant thus heat supplied in raising the temperature by ΔT is given as

$$Q = nC_p\Delta T$$

$$= n\left(\frac{7}{2}R\right)\Delta T \qquad [\text{As for } O_2 \text{ as being diatomic gas } C_p = \frac{7R}{2}]$$

$$Q = \frac{7}{2} \times 300 = 1050 \text{ J}$$

(iv) ADIABATIC PROCESS:

There is no exchange of heat between the system and the surrounding, but pressure volume and temperature of the system do change.

Equation for adiabatic process,

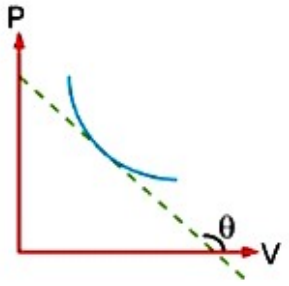
$PV^{\gamma} = C$ (constant)

or $T^{\gamma}P^{1-\gamma}$ = constant

or $TV^{\gamma-1}$ = constant

Indicator diagrams of adiabatic process

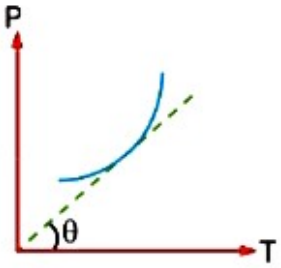
(a) P–V graph



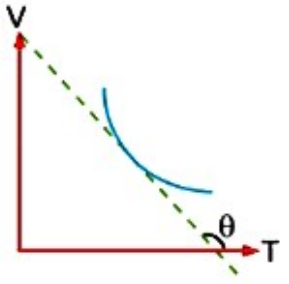
Slope of P–V curve

$$\tan\theta = \frac{dP}{dV} = -\gamma\left(\frac{P}{V}\right)$$

(b) P–T graph



Slope of P–T curve, $\tan\theta = \frac{dP}{dV} = \frac{\gamma}{(\gamma-1)}\left(\frac{P}{T}\right)$



(c) V–T graph

Slope of V–T curve, $\tan\theta = \frac{dV}{dT} = -\frac{1}{(\gamma-1)}\left(\frac{V}{T}\right)$

Work done in adiabatic process,

$$W = \int_{V_i}^{V_f} PdV = \int_{V_i}^{V_f} CV^{-\gamma}dV = \frac{[P_iV_i - P_fV_f]}{(\gamma-1)}$$

$$\Rightarrow W = \frac{nR(T_i - T_f)}{(\gamma-1)}$$

Work done by system is +ve if $T_i > T_f$
Work done by system is –ve if $T_i < T_f$
Change in internal energy,

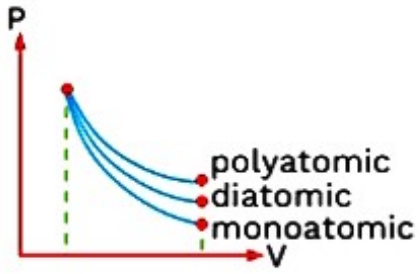
$$\Delta U = n\frac{f}{2}R\Delta T$$

By first law of thermodynamics,

$$Q = W + \Delta U$$

Since Q = 0, so $\Rightarrow W = -\Delta U$

This expression indicates that work is done by the system on the expense of its internal energy.
A sudden and quick process will be an adiabatic process due to lack of sufficient time for exchange of heat.
In adiabatic process, the slope of P–V curve for monoatomic gases and slope for diatomic is greater than polyatomic gases.



Slope of adiabatic P–V curve is greater than the slope of isothermal P–V curve.

Relation, $\left.\frac{dP}{dV}\right|_{adiabatic} = \gamma\left.\frac{dP}{dV}\right|_{isothermal}$

$$\Rightarrow \quad \gamma = \frac{\text{Slope of adiabatic P-V curve}}{\text{Slope of isothermal P-V curve}}$$

Q. **A motor tyre is pumped to a pressure of 3 atm. It suddenly bursts. Calculate the fall in temperature due to adiabatic expansion. The temperature of air before expansion is 27° C. Given $\gamma = 1.4$.**

Sol. We know that $T_2^{\gamma} P_2^{1-\gamma} = T_1^{\gamma} P_1^{1-\gamma} \Rightarrow \left[\frac{T_2}{T_1}\right]^{\gamma} = \left[\frac{P_1}{P_2}\right]^{1-\gamma} \Rightarrow \left[\frac{T_2}{300}\right]^{1.4} = \left[\frac{3}{1}\right]^{1-1.4}$

$\Rightarrow \left[\frac{T_2}{300}\right]^{1.4} = \left[\frac{1}{3}\right]^{0.4} \Rightarrow T_2 = 219.2\,K \quad \Rightarrow \Delta T = (300 - 219.2)K = 80.8\,K.$

Q. **A sample of diatomic gas with $\gamma = 1.5$ is compressed from a volume of 1600 cc to 400 cc adiabatically. The initial pressure of gas was 1.5×10^5 Pa. Find the final pressure and work done by the gas in the process.**

Sol. We know in an adiabatic process. pressure and volume of gas at its different states are related as

$$P_1 V_1^{\gamma} = P_2 V_2^{\gamma}$$

$$\text{or } P_2 = \left(\frac{V_1}{V_2}\right)^{\gamma} . P_1$$

$$= \left(\frac{1600}{400}\right)^{1.5} \times 1.5 \times 10^5 = (4)^{1.5} \times 1.50 \times 10^5$$

$$= 1.2 \times 10^6 \text{ Pa}$$

For an adiabatic process, work done by a gas is given as

$$W = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1}$$

$$= \frac{1.5 \times 10^5 \times 1600 \times 10^{-6} - 1.2 \times 10^6 \times 400 \times 10^{-6}}{1.5 - 1}$$

$$= \frac{240 - 480}{0.5} = -480 J$$

Here work by gas comes out a negative value, thus we can state that as gas is being compressed, work is done on the gas and so work done by gas is —480 J.

(v) POLYTROPIC PROCESS:

Molar heat capacity of a gas remains constant during polytropic process.

The four standard processes which are discussed earlier are the special cases of polytropic process.

By first law of thermodynamics,

$$\delta Q = \delta W + dU$$

$$\Rightarrow nCdT = PdV + n\frac{f}{2}RdT$$

$$\Rightarrow nCdT = PdV + \frac{n}{(\gamma - 1)}RdT$$

$$\Rightarrow C = \frac{RPdV}{RndT} + \frac{R}{(\gamma - 1)} \qquad ...(1)$$

By equation of state,

$PV = nRT$

Differentiated form,

$PdV + VdP = nRdT$

Put in (1),

$$\Rightarrow C = \frac{R}{(\gamma - 1)} + \frac{RPdV}{PdV + VdP} \qquad ...(2)$$

$$\Rightarrow \left[C - \frac{R}{(\gamma - 1)}\right] = \frac{RPdV}{PdV + VdP}$$

Put $\left[C - \frac{R}{(\gamma - 1)}\right] = K$ (constant),

$$\Rightarrow \frac{dV}{V}\left[\frac{K - R}{K}\right] + \frac{dP}{P} = 0$$

Integrate the equation,

$$\Rightarrow \left(\frac{K - R}{K}\right)\ell nV + \ell nP = C_1 \qquad \text{(constant)}$$

$$\Rightarrow P.V^{\left(\frac{K-R}{K}\right)} = \text{constant}$$

Where, $\frac{K - R}{K} = n$ (polytropic index)

and **$[PV^n = \text{constant}]$**

The above relation is known as polytropic equation. Differentiated form of above equation gives,

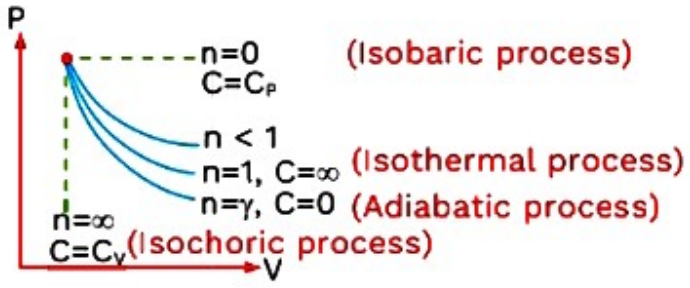
$$dP = -\frac{nP}{V}dV$$

Put in equation (2),

$$\Rightarrow C = \frac{R}{\gamma - 1} + \frac{RPdV}{PdV + V\left(\frac{-nP}{V}dV\right)}$$

$$\Rightarrow \left[C = \frac{R}{\gamma - 1} + \frac{R}{1-n} = C_v + \frac{R}{1-n}\right]$$

The above equation gives the molar heat capacity for a polytropic process.
Indicator diagram for polytropic process,

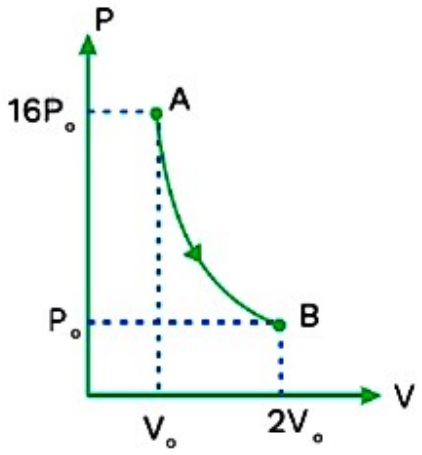


Work done in a polytropic process,

$W = n(C - C_v)\Delta T = \frac{P_1V_1 - P_2V_2}{n-1}$ Change in internal energy,

$\Delta U = nC_v\Delta T = n\frac{f}{2}R\Delta T$

Q. **Figure demonstrates a polytropic process (i.e., PV^n = constant) for an ideal gas. The work done by the gas in process AB will be:**



(1) $\frac{15}{2}P_0V_0$ **(2)** $\frac{14}{3}P_0V_0$

(3) $8P_0V_0$ **(4) Insufficient information**

Sol. For a polytropic process, $PV^n = constant \Rightarrow 16P_0V_0^n = P_0(2V_0)^n \Rightarrow n = 4$

$$\text{Work done} = \frac{P_1V_1 - P_2V_2}{n-1} = \frac{16P_0V_0 - P_0(2V_0)}{4-1} = \frac{14}{3}P_0V_0$$

Q. **One mole of an ideal gas with heat capacity at constant pressure C_p undergoes the process $T = T_0 + \alpha V$, T_0 and α are constant. Find:**
(a) heat capacity of the gas as a function of its volume,
(b) the amount of heat transferred to the gas if its volume is increased from V_1 to V_2.

Sol. The process equation for the thermodynamic process is given as

$$T = T_0 + \alpha V$$

From gas law, we have

$$P = \frac{RT}{V} = \frac{R}{V}(T_0 + \alpha V) \qquad \text{[As n = 1 mole]} \ldots\ldots\ldots\ldots\text{(i)}$$

(a) The heat capacity of a gas in a thermodynamic process is given as

$$C = C_v + \frac{RPdV}{PdV + VdP} \qquad \ldots\ldots\ldots\ldots\text{(ii)}$$

Differentiating equation —(i), we get

$$PdV + VdP = \alpha RdV \qquad \ldots\ldots\ldots\ldots\text{(iii)}$$

Now from equation —(i), (ii) and (iii), we have

$$C = C_v + \frac{R\left[\left(\frac{R}{V}\right)(T_0 + \alpha V)\right]dV}{\alpha RdV} = C_v + \frac{R}{\alpha V}(T_0 + \alpha V)$$

or $$= C_v + \frac{RT_0}{\alpha V} + R$$

$$= C_p + \frac{RT_0}{\alpha V}$$

(b) As volume of gas changes from V_1 to V_2, the corresponding temperatures of the gas are

$$T_1 = T_o + \alpha V_1$$

and $T_2 = T_o + \alpha V_2$

The amount of heat supplied to the gas can be given as

$$Q = \int dQ = \int_{T_1}^{T_2} CdT \qquad \text{[As n = 1 mole]}$$

or $$= \int_{T_1}^{T_2} \left[C_P + \frac{RT_o}{\alpha V} \right] dT$$

From the given relation we have

$$V = \frac{T - T_o}{\alpha}$$

Thus $$Q = \int_{T_1}^{T_2} \left[C_P + \frac{RT_o}{T - T_o} \right] dT$$

$$= C_P(T_2 - T_1) + RT_o \ln\left(\frac{T_2 - T_o}{T_1 - T_o} \right)$$

Q. **An ideal gas has a molar heat capacity C_v at constant volume. Find the molar heat capacity of this gas as a function of its volume V, if the gas undergoes the following processes:**

(a) $T = T_0 e^{\alpha V}$ **and** **(b)** $P = P_0 e^{\alpha V}$.

Sol. (a) The process equation for the thermodynamic process is given as

$$T = T_0 e^{\alpha v} \quad ..(i)$$

For a general thermodynamic process, we know the molar heat capacity is given by

$$C = C_v + \frac{RPdV}{PdV + VdP} \quad .....(ii)$$

From gas law we can relate pressure and volume of gas as

$P = \frac{nRT}{V}$ From equation — (i),

$$P = \frac{nRT_0}{V} e^{\alpha V} \quad \text{.....(iii)}$$

Differentiating this equation, we get

$$PdV + VdP = nRT_0 \alpha e^{\alpha V} dV \quad \text{.....(iv)}$$

From equation —(ii), (iii) and (iv), we get

$$C = C_v + \frac{R\left(\frac{nRT_0}{V} e^{\alpha V}\right) dV}{nRT_0 \alpha e^{\alpha V} dV}$$

or $$C = C_v + \frac{R}{\alpha V} \quad \text{.....(v)}$$

Equation —(v) gives the molar specific heat of the gas undergoing the given process and students should note that this molar specific heat is given as function of volume of gas thus this process is a non-polytropic process.

(b) In this case, the process equation is given as

$$P = P_0 e^{\alpha V} \quad \text{.....(vi)}$$

Differentiating, we get

$$dP = P_0 \alpha e^{\alpha V} dV \quad \text{.....(vii)}$$

If C is the molar specific heat of the gas in this process, then from equation —(ii), (vi) and (vii) we have

$$C = C_v + \frac{R(P_0 e^{\alpha V})dV}{(P_0 e^{\alpha V})dV + V(P_0 \alpha e^{\alpha V} dV)}$$

or $$C = C_v + \frac{R}{1 + \alpha v} \quad \text{.....(viii)}$$

Equation —(viii) gives the molar heat capacity of the gas in the given process and again we can say that this is also not a polytropic process.

(vi) FREE EXPANSION:

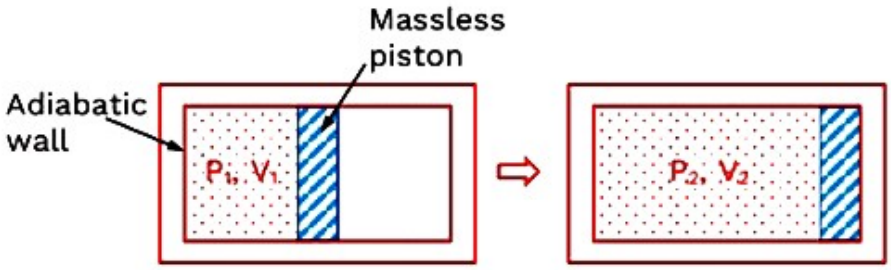
In free expansion, there is no heat exchange (Q = 0) between the system and surrounding. Work done by or on the system is zero (W = 0) and temperature of the system remains constant throughout the process,

$$Q = 0 ; \quad W = 0 ; \quad \Delta T = 0 ; \quad \Delta U = 0$$

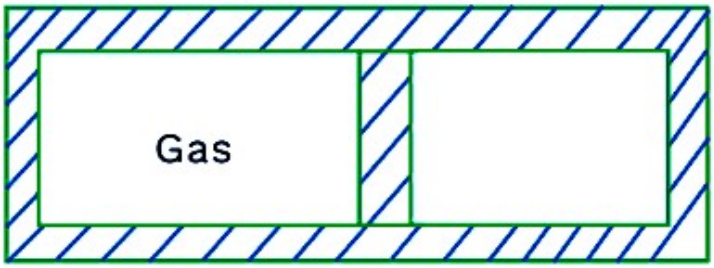
During free expansion, work done by gas is zero because there is nothing on which work can be done.

Boyle's law is valid as temperature remains constant,

$$P_1V_1 = P_2V_2$$



Q. **Figure shows a cylindrical container of volume V whose walls are adiabatic. Initially a light adiabatic piston divides the container in two equal parts as shown. In left part there are n moles of an ideal gas with adiabatic exponent γ filled at temperature T and in other part, there is vacuum. If the piston is released, the gas fills the whole container uniformly. Find the final pressure and temperature of gas. Now if the piston is slowly displaced externally back to its initial position, find the final pressure and temperature of gas.**



Sol. When the piston is released, the gas expands to fill the complete volume of container. As there is nothing in the other part of cylinder, this is the free expansion of gas hence no work is done by the gas and as container is thermally insulated from surrounding, the gas temperature remains constant and thus according to Boyle's Law as the volume of gas is doubled so its final pressure is reduced to half. Thus

$$P_i = \frac{nRT}{\left(\frac{V}{2}\right)}$$

Thus, we have $P_f = \frac{P_i}{2} = \frac{nRT}{V}$

Now if piston is displaced back to its initial position, this is adiabatic compression of gas in which gas volume decreases to half i.e. V/2. As we know in adiabatic process, pressure and volume of gas in different states are related as

$$P_1V_1^{\gamma} = P_2V_2^{\gamma}$$

Here $P_1 = P_f = \frac{nRT}{V}$, $V_1 = V$ and $V_2 = \frac{V}{2}$,

Thus, $P_2 = P_1\left(\frac{V_1}{V_2}\right)^{\gamma}$

$= P_1(2)^{\gamma}$

$= (2)^{\gamma}\frac{nRT}{V}$

Similarly in adiabatic process, gas volume and temperature in different states are related as

$$T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$$

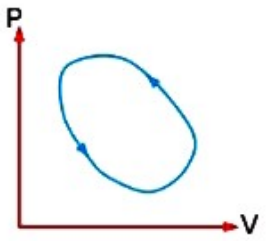
or $T_2 = T_1\left(\frac{V_1}{V_2}\right)^{\gamma-1}$

Here $T_1 = T, V_1 = V$ and $V_2 = \frac{V}{2}$ thus, $T_2 = T(2)^{\gamma-1}$.

(vii) CYCLIC PROCESS:

In cyclic process, initial and final thermodynamic states are same after undergoing a series of changes.

$P_f = P_i$, $V_f = V_i$, $T_f = T_i$



Work done in a cyclic process is equal to area enclosed under P–V diagram.

Change in internal energy $\Delta U = 0$,

So, $Q = W$

Clockwise P–V curve → Work done is +ve
Anticlockwise P–V curve → Work done is –ve

Efficiency of a Cyclic Process (η)

$$\eta = \frac{\text{Total mechanical work done by the gas}}{\text{Heat absorbed by the gas (Only + Ve)}}$$

Or

$$\eta = \frac{\text{Area under the cycle in P – V curve}}{\text{Heat injected into the system}}$$

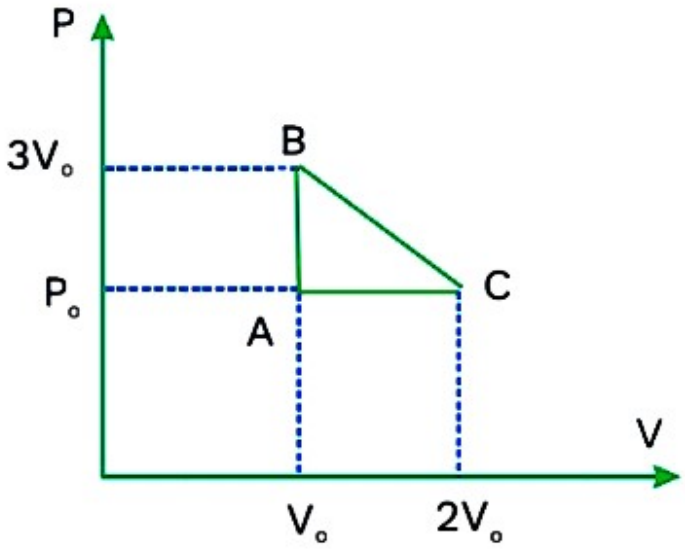
Q. **One mole of an ideal monatomic gas is taken round the cyclic process ABCA as shown in figure. Calculate**

(a) The work done by the gas.

(b) The heat rejected by the gas in the path CA and the heat absorbed by the gas in the path AB.

(c) The net heat absorbed by the gas in the path BC.

(d) The maximum temperature attained by the gas during the cycle.



Sol. (a) The work done by the gas is equal to the area under the closed curve. Thus, work done is cycle is:

$$W = \frac{1}{2}(2V_o - V_o)(3P_o - P_o)$$

$$= \frac{1}{2}V_o \times 2P_o$$

$$= P_oV_o$$

(b) Heat rejected in path CA is given as

$$Q_{CA} = nC_P\Delta T = nC_P(T_C - T_A)$$

$$= 1 \times \frac{5}{2}.R\left[\frac{P_o 2V_o}{R} - \frac{P_oV_o}{R}\right] = \frac{5}{2}P_oV_o$$

Heat absorbed in path AB

$$Q_{AC} = nC_V(T_B - T_A)$$

$$= 1 \times \frac{3}{2}R \times \left[\frac{3P_oV_o}{R} - \frac{P_oV_o}{R}\right] = 3P_oV_o$$

(c) For cycle ABC, we have
Heat supplied = work done by the gas

Or $$-\frac{5}{2}P_oV_o + 3P_oV_o + Q_{BC} = P_oV_o$$

Heat supplied in path BC is given by

$$Q_{BC} = P_oV_o + \frac{5}{2}P_oV_o - 3P_oV_o = \frac{P_oV_o}{2}$$

(d) We know that PV/T = constant. So, when PV is maximum, T is also maximum. PV is maximum for part BC. Hence temperature will be maximum between B and C.

Let equation of BC be as

$P = kV + k'$ satisfying both the points B and C.

For point B, $3P_0 = kV_0 + k'$

For point C, $P_0 = k(2V_0) + k'$

Solving these equations, we get

$$k = -2(P_0 / V_0) \quad \text{and} \quad k = 5P_0$$

So, the equation for line BC is

$$P = -\left(\frac{2P_0}{V_0}\right)V + 5P_0$$

Or $$\frac{RT}{V} = -\frac{2P_0V}{V_0} + 5P_0$$

Or $$T = \frac{P_0}{R}\left[5V - 2\frac{V^2}{V_0}\right] \quad \text{.....(i)}$$

For maximum T, dT/dV = 0

So, $$\frac{dT}{dV} = \frac{P_0}{R}\left[5 - \frac{4V}{V_0}\right] = 0$$

Hence $$5 - \frac{4V}{V_0} = 0 \text{ or } 5V_0 - 4V = 0$$

Or $$V = \frac{5}{4}V_0 \quad \text{.....(ii)}$$

Substituting the value of V from equation —(ii) in equation —(i), we get

$$T_{max} = \frac{P_0}{R}\left[5 \times \left(\frac{5}{4}V_0\right) - 2\left(\frac{5V_0}{4}\right)\frac{1}{V_0}\right]$$

$$= \frac{P_0}{R}\left[\frac{25V_0}{4} - \frac{25V_0}{8}\right] = \frac{P_0}{R} \times \frac{25V_0}{8} = \frac{25P_0V_0}{8R}$$

Q. **A cylinder containing 2 litres of air at 2 atmospheric pressure and temperature 200 K, is subjected to the following cyclic process in four different stages. It is (a) heated at constant pressure to 600 K, (b) cooled at constant volume to 300 K, (c) cooled at constant pressure to 100 K and (d) heated at constant volume to 200 K. Depict the complete cycle on a P—V diagram showing the values of P and V at the end of each stage.**

Sol. Let initially, volume V_0 = 2 litres, P_0 = 2 atm, and temperature T_0 = 200 K.

(a) Since pressure is constant, P_1 = 2 atm and T_1 = 600 K.

Applying Charles's law, $V \propto T \Rightarrow \frac{V_1}{V_0} = \frac{T_1}{T_0}$

or $\frac{V_1}{2\,\text{litres}} = \frac{600\,K}{200\,K} \Rightarrow V_1 = 6\,\text{litres}$

(b) Since volume is constant, $V_2 = 6$ litres and $T_2 = 300$ K.

Applying pressure law, $P \propto T \Rightarrow \frac{P_2}{P_1} = \frac{T_2}{T_1}$

Or $\frac{P_2}{2\,\text{atm}} = \frac{300\,K}{600\,K} \Rightarrow P_2 = 1\,\text{atm}$

(c) Since, pressure is constant, so $P_3 = 1$ atm and $T_3 = 100\,K$.

Applying Charles's law, $V \propto T \Rightarrow \frac{V_3}{V_2} = \frac{T_3}{T_2}$

or $\frac{V_3}{6\text{ litres}} = \frac{100\,K}{300\,K} \Rightarrow V_3 = 2\,\text{litres}$

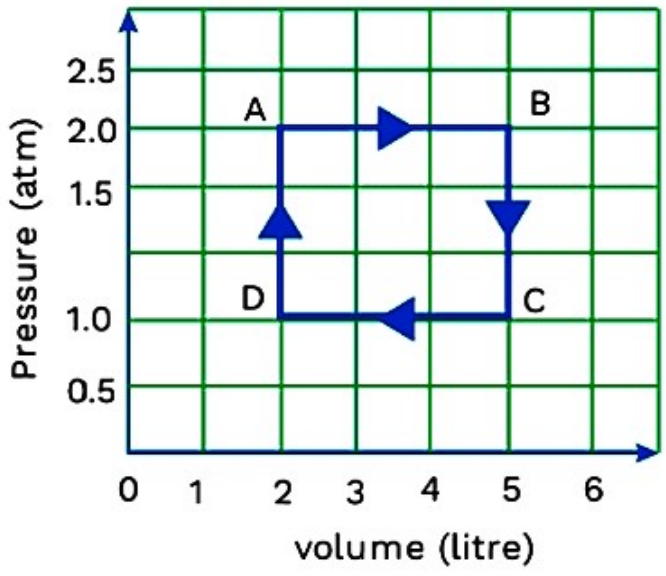
(d) Finally, since volume remains constant,

$V_4 = 2$ litres and $T_4 = 200\,K$

Applying pressure law, $P \propto T \Rightarrow \frac{P_4}{P_3} = \frac{T_4}{T_3}$

or $\frac{P_4}{1\text{ atm}} = \frac{200\,K}{100\,K} \Rightarrow P_4 = 2\,\text{atm}$

Thus, comparing P_0, V_0 and T_0 with P_4, V_4 and T_4, it is evident that, the air is brought back to its initial state. All these processes are shown in given figure.



Second Law of Thermodynamics

Reversible and Irreversible Processes:

Reversible Process :- When the system is in thermal equilibrium with itself and with its surrounding then any change of state that takes place can be reversed by making a very small change in the conditions, such processes are known as reversible processes or equilibrium processes.
Reversibility is an ideal concept that can never be realized in practical world.

Quasi-Static Process :- To make a process nearly reversible, the process is carried out very slowly so that it can be considered to be in thermal equilibrium at every state, such process is known as quasi-static process.

Irreversible Process/Non-Equilibrium Process :-

The process which is not reversible is the irreversible process.
Thermodynamic processes that occur in nature, free expansion of gas or conversion of work to heat are all irreversible process.

Second law of Thermodynamics

Second law of thermodynamics is explained by two individual statements. These are stated as

(i) **Kelvin-Plank Statement:**

"It is impossible for any system to absorb heat and convert it completely into mechanical work leaving the system in its initial state from which the process was started."
In simple words, it is impossible for a heat engine to produce a net work in a complete cycle if it exchanges heat only with a single reservoir.

(ii) **Clausius Statement: -**

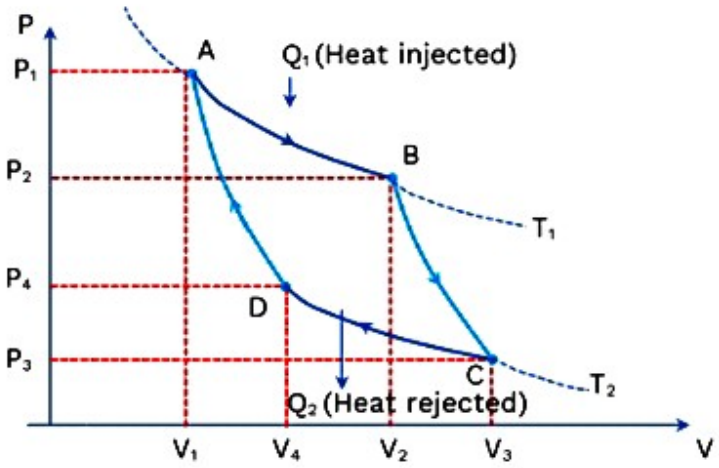
"It is impossible for any process to have only the heat flow from a colder to hotter body without any involvement of an internal energy."
Heat pump and refrigerator works on Clausius's statement.
The second law of thermodynamics tells us about the feasibility of any process by stating that the entropy (S) of the system going through a process will either remain constant (reversible process) or increase (spontaneously occurring process).

$$\Delta S_{universe} \geq 0$$

CARNOT CYCLE:

It is reversible cycle with four operations in succession. These are:
1. Isothermal expansion of working substance (gas), A→B.
2. Adiabatic expansion, B → C

3. Isothermal compression, C → D
4. Adiabatic compression to its initial state, D → A



Isothermal Expansion (A→B)

$$Q_1 = W_{AB} = nRT_1 \ell n\left(\frac{V_2}{V_1}\right)$$

Adiabatic Expansion (B → C)

$$W_{BC} = \frac{nR}{(V-1)}(T_1 - T_2); \qquad T_1V_2^{\gamma-1} = T_2V_3^{\gamma-1} \qquad ...(1)$$

Isothermal Compression (C → D)

$$Q_2 = W_{CD} = nRT_2 \ell n\left(\frac{V_4}{V_3}\right)$$

Adiabatic Compression (D → A)

$$W_{DA} = \frac{nR}{(\gamma-1)}(T_2 - T_1); \qquad T_1V_1^{\gamma-1} = T_2V_4^{\gamma-1} \qquad ...(2)$$

By equation (1) and (2)

$$\frac{V_2}{V_1} = \frac{V_3}{V_4}$$

Efficiency of a Carnot Cycle = $1 - \left|\frac{Q_2}{Q_1}\right| = 1 - \frac{T_2}{T_1}$

The efficiency for the Carnot engine can never be 100% because T_2 cannot be zero.
No irreversible engine can have efficiency greater than Carnot reversible engine working between same hot and cold reservoir.

Q. **The temperatures T_1 and T_2 of the two heat reservoirs in an ideal Carnot engine are 1500° C and 500° C respectively. Which of the following: increasing T_1 by 100° C or decreasing T_2 by 100° C would result in a greater improvement in the efficiency of the engine?**

Sol. $T_1 = 1500^\circ C = (1500 + 273)K = 1773K$ and $T_2 = 500^\circ C = (500 + 273)K = 773K$.

The efficiency of a Carnot's engine $\eta = 1 - \frac{T_2}{T_1}$

When the temperature of the source is increased by 100° C, keeping T_2 unchanged, the new temperature of the source is $T'_1 = 1500 + 100 = 1600^\circ C = 1873K$.

The efficiency becomes $\eta' = 1 - \frac{T_2}{T_1} = 1 - \frac{773}{1873} = 0.59$

On the other hand, if the temperature of the sink is decreased by 100° C, keeping T_1 unchanged, the new temperature of the sink is $T'_2 = 500 - 100 = 400^\circ C = 673K$.

The efficiency now becomes

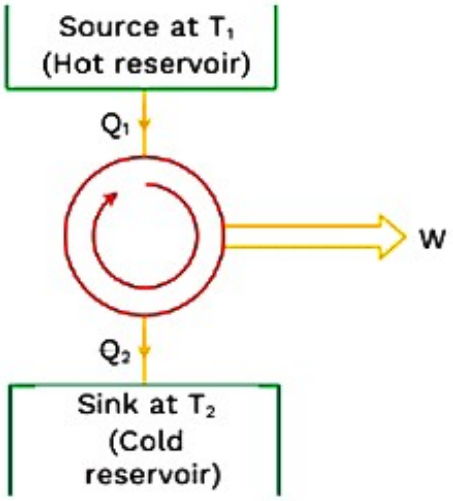
$$\eta'' = 1 - \frac{T_2'}{T_1} = 1 - \frac{673}{1773} = 0.62$$

Since η'' is greater than η', decreasing the temperature of the sink by 100° C results in a greater efficiency than increasing the temperature of the source by 100° C.

HEAT ENGINE (H.E.):

It is a thermodynamic system undergoing a clockwise cyclic process.

A device which absorbs heat from a source at relatively high temperature by using a working substance, performs some mechanical work and rejects some heat at relatively lower temperature is a heat engine.



By conservation of energy
Work done by H.E., $W = Q_1 - Q_2$
Efficiency of heat engine (η)

$$\eta = \frac{\text{Work done by Heat Engine}}{\text{Heat taken from source}} = \frac{Q_1 - Q_2}{Q_1}$$

$$\Rightarrow \quad \eta = 1 - \frac{Q_2}{Q_1} = 1 - \frac{T_2}{T_1}$$

Q. **A heat engine operates between a cold reservoir at temperature $T_2 = 3000$ K and a hot reservoir at temperature T_1. It takes 200 J of heat from the hot reservoir and delivers 120 J of heat the cold reservoir in cycle. What could be the minimum temperature of hot reservoir?**

Sol. Work done by the engine in a cycle is $W = 200 - 120 = 80$ J.

$$\eta = \frac{W}{Q} = \frac{80}{200} = 0.4$$

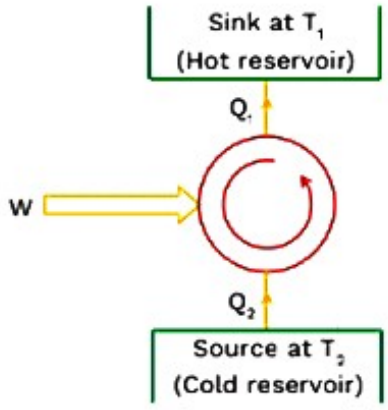
From Carnot's Theorem $0.4 \le 1 - \frac{T_2}{T_1} = 1 - \frac{300}{T_1} \Rightarrow \frac{300}{T_1} K \le 0.6 \Rightarrow T_1 \ge \frac{300}{0.6} \Rightarrow T_1 \ge 500$

REFRIGERATOR AND HEAT PUMP:

Working principle is same for both refrigerator and heat pumps as well.
The only difference between refrigerator and heat pump is the desired output.

Working Principle:-

It is a thermodynamic system undergoes an anti-clockwise cyclic process.
It extracts heat from a cold reservoir and reject it to a hot reservoir on the expense of mechanical energy supplied to it by an external agent.



By conservation of energy

$$W = Q_1 - Q_2$$

Output :

In case of refrigerator, the desired output is the heat extracted from cold reservoir (Q_2). The coefficient of performance

$$(COP)_R = \frac{\text{Heat extracted from cold reservoir}}{\text{Work done on refrigerator}} \qquad \left[(COP)_R = \frac{Q_2}{Q_1 - Q_2}\right]$$

In case of heat pump, the desired output is the heat rejected to a hot reservoir (Q_1). The coefficient of performance $(COP)_H$.

$$(COP)_H = \frac{\text{Heat rejected to hot reservoir}}{\text{Work done on heat pump}}$$

$$\left[(COP)_H = \frac{Q_1}{Q_1 - Q_2}\right]$$

Relation between $(COP)_H$ and $(COP)_R$

$$(COP)_H = \frac{Q_1}{Q_1 - Q_2} = \frac{Q_1 - Q_2}{Q_1 - Q_2} + \frac{Q_2}{Q_1 - Q_2}$$

$$\Rightarrow [(COP)_H = 1 + (COP)_R]$$

Q. **A refrigerator takes heat from water at 0°C and transfers it to room at 27° C. If 100 kg of water is converted in ice at 0° C, then calculate the work done. (Latent heat of ice is 3.4×10^5 J/kg)**

Sol. Coefficient of performance (COP) $= \frac{T_2}{T_1 - T_2} = \frac{273}{300 - 273} = \frac{273}{27}$

$$W = \frac{Q_2}{COP} = \frac{mL}{COP} = \frac{100 \times 3.4 \times 10^5}{273/27} = \frac{100 \times 3.4 \times 10^5 \times 27}{273} = 3.36 \times 10^7 \text{ J}$$